On-Device Small Language Models for Privacy-Preserving University Administration: A Benchmark on Commodity Hardware

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Abstract

Universities want to automate routine administrative text work, but sending student records to commercial cloud models raises disclosure questions under FERPA. Running small language models (SLMs) on hardware the institution already owns avoids that disclosure, if the models are fast enough and good enough. We benchmark two open-weight SLMs, Llama 3.2 3B and Qwen 2.5 7B, each at two quantization levels (`Q4_K_M` and `Q8_0`), on a Windows laptop (with and without its GPU) and an Apple M4 Mac. The four tasks are drawn from university operations: extracting structured fields from syllabi (24 documents), summarizing policies without losing deadlines or thresholds (16 policies, 52 required constraints), routing helpdesk tickets into eight categories (64 tickets), and drafting replies to students (16 inquiries, rated blind by two people). On the laptop's CPU alone, decode throughput ranged from 9.4 to 26.1 tokens/s. Qwen 2.5 7B (`Q4_K_M`) produced schema-valid JSON for 100% of syllabi versus 12% for Llama 3.2 3B, and classified 88% of tickets correctly versus 56%. Human raters judged that **[TBD]**% and **[TBD]**% of drafted replies, respectively, needed a substantive edit before sending. The 7B model at 4-bit quantization met the structural requirements of every extraction and summarization item, quantization level made no significant difference to any pass/fail metric, and even without a GPU a reply was drafted in about 22.1 s; human review remains necessary because both models occasionally invented contact details or altered a policy's conditions. We release the harness, data, and raw outputs.

Keywords

small language models, on-device inference, quantization, higher education, FERPA, accessibility, benchmarking

1 Introduction

Much of a university's administrative workload is text: helpdesk tickets that must be routed, syllabi that must be checked for required statements, policies that must be explained to students in plain language, and e-mails that must be answered. Large language models handle such text well, but the most capable ones run in commercial clouds. Sending student information to a cloud provider is a disclosure that FERPA [16] restricts, and institutions increasingly limit which external AI services staff may use [4].

A practical alternative is to run a *small* language model (SLM), here meaning an open-weight model of at most about 8B parameters, on a computer the institution already owns. Quantization makes such models fit in ordinary office hardware [10, 11]. Whether they are good *enough*, and fast *enough*, for specific administrative tasks is an empirical question that general-purpose benchmarks do not answer. Structured output that a database will ingest has to be valid every time, a policy summary that drops a deadline is worse than no summary, and a reply that invents a phone number creates work instead of saving it.

We ask four questions:

- **RQ1** How fast are 3B and 7B SLMs on an ordinary laptop, with and without a GPU, and on Apple Silicon, and how much memory do they need?
- **RQ2** How well do they perform four administrative tasks, and how much does model size or quantization level change that?
- **RQ3** How sensitive is task performance to the sampling temperature?
- **RQ4** How often does a human have to substantively correct their output before it can be used?

Our contributions are (1) a four-task benchmark built around concrete failure criteria (schema validity, retention of every deadline and threshold, exact label match, invented contact details), (2) a measurement method for interactive latency that records every token and separates model memory from whole-system memory, and (3) results for four model configurations on two machines, with uncertainty intervals and a blind human evaluation. Code, data, prompts and every raw model output are public.[1]

2 Background and Related Work

Regulatory context. FERPA prohibits disclosing personally identifiable information from education records without consent, outside defined exceptions [16]. Running inference locally removes one disclosure path, transmission to a model provider; it does not by itself make a workflow compliant, which still depends on access control, logging, and retention on the local machine. ADA Title II requires public universities to make web content and course materials accessible; the Department of Justice's 2024 rule adopted

[1] https://github.com/MohdIliyas/local-slm-highered-benchmark

WCAG 2.1 Level AA as the technical standard [17, 20], and an April 2026 interim final rule moved the first compliance date to April 26, 2027 for entities serving populations of 50,000 or more [18]. Accessibility work is largely text work: checking that syllabi contain an accommodation statement and summarizing remediation timelines without dropping them are two of our tasks.

Quantized local inference. Husom et al. [10] measured energy, latency and accuracy for 28 quantized models from the Ollama library on a Raspberry Pi 4, showing the trade-offs quantization creates on very constrained devices. Kurt [11] compared llama.cpp quantization formats on Llama-3.1-8B-Instruct across standard benchmarks, perplexity, and CPU throughput. Both evaluate general capability; we evaluate task-specific pass/fail criteria that determine whether an output is usable in an administrative pipeline.

Local models in regulated domains. Bhetwal et al. [1] benchmarked local models for natural-language-to-SQL querying of pharmaceutical manufacturing data on consumer hardware and found that human oversight remained necessary. Dong and Liebowitz [4] studied how much student data an advising assistant exposes to a third-party model provider and how to minimize it; on-device inference is the limiting case in which nothing is exposed to a provider.

Very large models on consumer hardware. Colibri [6] runs GLM-5.2, a 744B-parameter mixture-of-experts LLM, on a machine with roughly 25 GB of RAM by keeping the dense layers resident and streaming routed experts from an NVMe SSD. Its developers and users report decode speeds on the order of 0.05 to about 1 token/s on consumer machines, depending on disk bandwidth and cache state. We return to this alternative in Section 5.

3 Method

3.1 Hardware and software

Table 1 lists both machines. All inference runs through Ollama [14] (llama.cpp backend [7]) with default thread settings, a 4,096-token context, and one request at a time, the single-user setting of a staff member's computer. The Windows laptop has a discrete GPU, which Ollama uses by default; we therefore measure it twice, once with Ollama's default GPU offload and once with offload disabled (`num_gpu=0`), which stands in for the many office machines without a usable GPU. Quality results are reported from the default configuration; the CPU-only run is used for speed. Greedy decoding is reproducible within a configuration: a complete rerun with GPU offload produced byte-identical outputs for all 480 of 480 items. Across configurations, 372 of 480 outputs were identical between the GPU and CPU-only runs; the rest, mostly long reply drafts, diverge after small floating-point differences change one token, and the headline metrics differ by at most about six percentage points.

3.2 Models

We evaluate Llama 3.2 3B Instruct [8] and Qwen 2.5 7B Instruct [15], each in the `Q4_K_M` and `Q8_0` GGUF formats published in the Ollama library, giving four configurations. Each configuration is pulled by its exact tag, and the harness records each model's content digest.

Table 1: Test machines (recorded automatically by the harness).

	Windows laptop	Mac
CPU	Intel Core 7 240H	**[TBD]**
GPU	NVIDIA GeForce RTX 5060 Laptop GPU, 8 GB	integrated (Metal)
RAM (GB)	31.7	**[TBD]**
OS	Windows-11-10.0.26300-SP0	**[TBD]**
Ollama	0.40.6	**[TBD]**

3.3 Tasks and data

All task data are synthetic and were written for this study; none comes from real students, so the dataset can be released in full. Items were written to resemble real university material in vocabulary, length and format variation, and were reviewed by the authors, two of whom work in university online-learning support. [Note: Confirm this sentence describes the review you actually did.]

T1: Syllabus extraction (24 syllabi). The model must return JSON with the course code, instructor, office hours, grading breakdown (with numeric weights), and whether the syllabus contains a disability-accommodation statement. Syllabi vary in layout: percentages, the word "percent", point totals that must be converted to weights, dot-leader tables, and distractor names such as a lab coordinator. Eight of the 24 lack an accommodation statement, so that answering "true" every time is penalized.

T2: Policy summarization (16 policies). The model must summarize a policy in exactly three bullet points with no introduction, preserving every concrete constraint. Each policy lists its required constraints (52 in total), such as "3 business days" or "WCAG 2.1 Level AA".

T3: Ticket classification (64 tickets, 8 classes). The model must return exactly one of eight category names. There are eight tickets per class. No ticket contains its own category name, and several are deliberately ambiguous, for example a password problem that only appears during an exam.

T4: Reply drafting (16 inquiries). The model drafts an e-mail reply with a subject line and a fixed sign-off, and is told not to invent dates, links, or contact names. Two inquiries per ticket class are used.

3.4 Automatic metrics

T1. *Schema validity*: the output parses as JSON and validates against the schema, including numeric types; a surrounding Markdown code fence or sentence is stripped first, and its presence is reported separately because a strict parser would reject it. *Field accuracy*: course code, instructor name, accommodation flag, and grading weights match the gold labels; we accept the weights either itemized or merged ("Exams 45%" for three 15% exams). Scoring field content separately from schema validity distinguishes a model that extracts the right numbers but writes `"45%"` as a string from one that extracts the wrong numbers. **T2.** *Format compliance*: exactly three bullets with nothing before or after them. *Constraint retention*: the share of required constraints present in the summary, matched as

whole phrases after normalizing number words and percent signs, so "five business days" matches "5 business days" but "15 business days" does not. After inspecting the first run we added inflected forms ("automatically convert to an F") for one constraint whose verb form correct summaries changed; no constraint was removed or weakened. We also report ROUGE [12] and BERTScore [21] against author-written reference summaries, as secondary measures. **T3.** *Strict accuracy*: the output, after trimming whitespace and punctuation, equals the gold label. Any other output counts as wrong. We also report a lenient variant, which credits an output that mentions exactly one label, and macro-F1. **T4.** Presence of the subject line and sign-off, and whether the draft invents a URL, e-mail address, or phone number.

3.5 Systems measurements

Every request is streamed. *Time to first token* (TTFT) is measured by the client from just before the request is sent until the first chunk containing text arrives. *Inter-token latency* (ITL) is the gap between consecutive text chunks; we confirm that the server emits about one token per chunk. *Decode throughput* is Ollama's own count of generated tokens divided by its decode time, which excludes model loading and prompt processing. *Memory* is the size Ollama reports for the loaded model; we also log peak resident memory summed over Ollama's processes, never whole-system memory. Before each configuration, all models are unloaded and the first, cold load time is recorded separately. Latency statistics come from the same distinct prompts as the quality results, not from repeating one prompt, which Ollama's prompt cache would make artificially fast.

3.6 Human evaluation

For the policy summaries and drafted replies of all four configurations ([TBD] outputs), two raters scored each output on a 1–5 usefulness scale and marked whether it needed a substantive edit and whether it contained a factual error. Raters worked from a written rubric, saw outputs in a fixed random order, and never saw model names. The *Human Intervention Rate* (HIR) of a configuration is the share of its outputs marked as needing a substantive edit. We report agreement as Cohen's κ [3], quadratically weighted for the 1–5 scale. A third author adjudicated disagreements blind. Because the raters are also authors, they were not blind to the study's hypotheses; we discuss this in Section 6.

3.7 Statistics

Proportions are reported with Wilson 95% intervals [19], and means and macro-F1 with percentile bootstrap intervals (2,000 resamples) [5]. Paired comparisons between configurations on the same items use the exact McNemar test [13] with Holm correction across all comparisons [9]. Main results use greedy decoding ($T = 0$, fixed seed). The temperature study repeats T1–T3 at $T = 0.2$ and $T = 0.7$ with three seeds each.

4 Results

4.1 Speed and memory (RQ1)

Table 2 summarizes the systems measurements. With GPU offload, the three configurations that fit in the laptop's 8 GB of GPU memory decoded at 56.8 to 100.3 tokens/s and returned the first token of a policy summary in 0.05 to 0.08 s. Qwen 2.5 7B at `Q8_0` (8.0 GB) did not fit; Ollama split it between GPU and CPU and its decode rate fell to 20.9 tokens/s. Without the GPU, decode ranged from 9.4 tokens/s (Qwen, `Q8_0`) to 26.1 tokens/s (Llama, `Q4_K_M`), and first-token latency rose to 0.61–2.86 s. Adults read silently at about 240 words per minute [2], roughly five to six tokens per second, so every configuration generates text faster than a person reads it. For a whole reply the wait is what matters: the median draft took 5.5 s with Qwen `Q4_K_M` on the GPU and 22.1 s on the CPU alone (33.1 s at `Q8_0`). Generation did not stall: the 99th-percentile inter-token latency was at most 2.1 times the median with GPU offload and within 1.1 times on the CPU alone, where timing was steadiest.

4.2 Task quality (RQ2)

Table 3 reports the four tasks on the Intel workstation. Llama 3.2 3B (`Q4_K_M`) produced schema-valid output for 12% of syllabi; in 88% it wrote a grading weight as a string. Its field accuracy, which ignores types, was 79%. Qwen 2.5 7B wrapped the JSON in a Markdown code fence in 83% of outputs, which a strict parser would reject; its schema validity is measured after removing the fence. The corresponding figures for Qwen 2.5 7B are 100%, 0%, and 89%. For policy summaries, 0% and 100% of outputs met the format rule, and all required constraints were kept in 69% and 88% of summaries, respectively. Strict classification accuracy was 56% and 88%; lenient matching raises these to 56% and 88%. The gap between the two is the share of answers a downstream router could not parse. Drafts containing invented contact details made up 12% and 6%. Model size mattered more than anything else we varied. Of the 12 paired comparisons between a 3B and a 7B configuration on schema validity, format compliance and classification accuracy, 11 remained significant after Holm correction; the exception was classification accuracy between the two `Q8_0` models ($p_{\text{Holm}} = 0.058$). [Note: Once the Apple M4 results are in, say whether they agree.]

Quantization. With GPU offload, Qwen 2.5 7B at `Q8_0` (8.0 GB) did not fit entirely in GPU memory, and Ollama placed part of the model on the CPU; its decode speed (20.9 versus 56.8 tokens/s at `Q4_K_M`) therefore reflects memory spill-over as much as quantization, and the CPU-only comparison is the clean one. Moving from `Q4_K_M` to `Q8_0` changed no pass/fail metric significantly after correction (smallest $p_{\text{Holm}} = 0.09$). The largest effect was on Llama's schema validity: 8 syllabi became valid at `Q8_0` and 0 became invalid (uncorrected $p = 0.008$), because the higher-precision model wrote weights as strings less often (54% versus 88%). For Qwen, `Q8_0` raised field accuracy from 89% to 98%, but it needs 1.7 to 1.8 times the memory and, on the CPU, decodes at 9.4 rather than 12.7 tokens/s. For these tasks the 4-bit format is the better default.

4.3 Sensitivity to temperature (RQ3)

Table 5 shows how performance and run-to-run agreement change with sampling temperature. Sampling barely affected Qwen: at

Table 2: Systems measurements. Decode throughput is mean ± s.d. across requests. TTFT is the median for policy prompts. Inter-token latency is pooled over all extraction, policy and drafting requests.

		Size	Decode	Prefill	TTFT	Inter-token latency (ms)		
Model	Quant.	(GB)	(tok/s)	(tok/s)	p50 (s)	p50	p95	p99
Windows laptop, GPU offload (Ollama default)								
Llama 3.2 3B	Q4_K_M	2.4	100.3 ± 7.6	8573	0.06	9.5	13.7	20.0
Llama 3.2 3B	Q8_0	3.7	80.5 ± 6.4	7909	0.05	11.8	15.7	22.8
Qwen 2.5 7B	Q4_K_M	4.4	56.8 ± 2.1	4781	0.08	17.9	19.6	24.9
Qwen 2.5 7B	Q8_0	8.0	20.9 ± 0.3	699	0.19	47.9	51.7	52.7
Windows laptop, CPU only (num_gpu=0)								
Llama 3.2 3B	Q4_K_M	2.4	26.1 ± 0.6	545	0.61	38.9	40.6	42.1
Llama 3.2 3B	Q8_0	3.7	19.5 ± 0.2	274	1.24	51.8	53.8	54.9
Qwen 2.5 7B	Q4_K_M	4.7	12.7 ± 0.2	258	1.34	79.3	81.7	84.8
Qwen 2.5 7B	Q8_0	7.9	9.4 ± 0.1	125	2.86	107.9	111.3	115.6
Apple M4 (Metal)								
Llama 3.2 3B	Q4_K_M	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Llama 3.2 3B	Q8_0	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Qwen 2.5 7B	Q4_K_M	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Qwen 2.5 7B	Q8_0	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Table 3: Task quality on the Intel workstation, $T = 0$. Brackets are 95% Wilson intervals. Str. wt.: share of extractions with a grading weight written as a string. Constr.: mean share of required constraints retained.

		Extraction			Policy summary		Classification	
Model	Quant.	Schema %	Fields %	Str. wt. %	Format %	Constr. %	Acc. %	Macro-F1
Llama 3.2 3B	Q4_K_M	12 [4, 31]	79	88	0 [0, 19]	89	56 [44, 68]	0.50
Llama 3.2 3B	Q8_0	46 [28, 65]	83	54	0 [0, 19]	90	66 [53, 76]	0.61
Qwen 2.5 7B	Q4_K_M	100 [86, 100]	89	0	100 [81, 100]	96	88 [77, 94]	0.87
Qwen 2.5 7B	Q8_0	96 [80, 99]	98	4	88 [64, 97]	94	86 [75, 92]	0.85

Table 4: Reference-based summary metrics (secondary).

Model	Quant.	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Llama 3.2 3B	Q4_K_M	0.693	0.470	0.601	[TBD]
Llama 3.2 3B	Q8_0	0.662	0.418	0.549	[TBD]
Qwen 2.5 7B	Q4_K_M	0.697	0.456	0.592	[TBD]
Qwen 2.5 7B	Q8_0	0.695	0.454	0.605	[TBD]

Table 5: Effect of sampling temperature (Q4_K_M). For $T > 0$, values average three seeds per item. Seed agreement is the share of items where all three seeds gave the same pass/fail outcome.

		Schema %	Format %	Acc. %	Seed agreement %	
Model	T				Schema	Acc.
Llama 3.2 3B	0.0	12	0	56	–	–
	0.2	17	0	56	83	88
	0.7	31	0	57	42	73
Qwen 2.5 7B	0.0	100	100	88	–	–
	0.2	100	100	86	100	97
	0.7	99	98	87	96	95

$T = 0.2$ its three seeds gave the same schema-validity outcome on 100% of syllabi, and even at $T = 0.7$ they agreed on 96% and accuracy stayed at 87%. The one cost was constraint retention: the share of summaries keeping every constraint fell from 88% at $T = 0$ to 77% at $T = 0.7$. Llama was unstable: at $T = 0.7$ its seeds agreed on schema validity for only 42% of syllabi. Its schema-valid rate rose from 12% to 31%, but only because sampling sometimes avoided the string-typed weights; that is luck, not reliability. Structured administrative tasks should use greedy decoding.

4.4 Human evaluation (RQ4)

Table 6 reports the blind ratings of drafted replies. Agreement between raters was κ_w = [TBD] for the quality score and κ = [TBD] for the substantive-edit judgment. [Note: Interpret HIR differences; mention the most common reasons from the raters' notes.]

4.5 Failure modes

Four patterns account for most errors. *Preambles*: Llama began every policy summary with a sentence such as "Here is a summary of the policy in three concise bullet points:", despite an explicit instruction not to; a system that publishes summaries automatically would have to strip it. *Changed conditions*: the finals policy lets a student

Table 6: Blind human ratings of drafted replies, averaged over raters (range across raters in parentheses).

Model	Quant.	Quality (1–5)	HIR %	Factual err. %	Invented contact %
Llama 3.2 3B	Q4_K_M	[TBD]	[TBD]	[TBD]	12
Llama 3.2 3B	Q8_0	[TBD]	[TBD]	[TBD]	0
Qwen 2.5 7B	Q4_K_M	[TBD]	[TBD]	[TBD]	6
Qwen 2.5 7B	Q8_0	[TBD]	[TBD]	[TBD]	0

reschedule if two exams fall at the same *time*; Qwen Q4_K_M summarized this as two finals on the same *day*, Llama as "two or more" exams on the same day, and Qwen Q8_0 dropped the condition altogether. Each changes who is eligible, and none would be caught by a fluency metric. *Invented contacts*: Llama Q4_K_M invented addresses such as disability.services@university.edu and matching web links, and Qwen Q4_K_M once invented a support e-mail address, despite being told not to. *Field boundaries*: Qwen Q4_K_M put the course title into the course-code field ("ENG 101: College Writing I") for 5 of 24 syllabi, while Llama's misclassified tickets most often fell back to "General Inquiries".

5 Discussion

What local inference does and does not secure. Running the model on a university-owned machine means prompts and outputs never reach a model provider, which removes the disclosure question that cloud APIs raise. It does not address who can access the machine, where outputs are stored, or how long logs are kept; those remain institutional controls.

Labor. The HIR translates into staff time only under assumptions about how long tasks take. With a manual draft time of 6.0 minutes, 1.5 minutes to review an acceptable draft, and 4.0 minutes to fix a flagged one, the time saved per ticket is

$$\Delta t = t_{\text{manual}} - t_{\text{review}} - \text{HIR} \cdot t_{\text{fix}}. \tag{1}$$

These times are illustrative assumptions, not measurements, and the conclusion should be read as a sensitivity analysis: the break-even HIR is $(t_{\text{manual}} - t_{\text{review}})/t_{\text{fix}}$. [Note: Measure the three times (for example, time yourselves on five tickets) and update configs/experiments.json, or keep this explicitly hypothetical.]

Dense SLMs versus disk-streamed MoE. Disk-streaming engines such as Colibri make a frontier-scale mixture-of-experts LLM runnable on a desktop, but at the reported speeds of roughly 0.05 to 1 token/s, a 200-token reply takes minutes rather than seconds. That rules them out for interactive helpdesk work but not for overnight batch jobs, such as auditing a semester's syllabi, where quality matters more than latency. We did not evaluate such a model; comparing its task quality with the dense SLMs here, at matched wall-clock budgets, is a natural next study.

6 Limitations

The data are synthetic and small (24, 16, 64 and 16 items). They avoid privacy problems, but real tickets are messier, and the confidence intervals are correspondingly wide. The reference summaries and gold labels were written by the authors, and the human raters are authors. We test two model families, two quantization formats, one inference server, and single-user load on two machines, one of them in two configurations; batch throughput, energy use, and long-context documents are out of scope. Results may change with Ollama or model updates, which is why the harness records versions and model digests.

7 Conclusion

On a mid-range Windows laptop, open-weight SLMs are fast enough for administrative text work with a GPU and still usable without one, at roughly 13 to 26 tokens per second on the CPU at 4-bit quantization. The 7B model met the structural requirements of nearly every item, while the 3B model's failures were mostly formatting errors that downstream code would have to repair; 4-bit quantization cost nothing measurable in quality, and greedy decoding should be the default. [Note: Add the human-evaluation finding (HIR) and the Apple M4 result in one sentence each.]

Data and Code Availability

The harness, task data, prompts, raw outputs, rating sheets and the scripts that generate every table in this paper are available at https://github.com/MohdIliyas/local-slm-highered-benchmark.

Ethics Statement

No real student data were used. All task items are synthetic. [Note: Ask the CSU IRB whether the internal human-rating study, in which the authors rate model outputs, needs a determination; record the answer here.]

Use of Generative AI

Portions of the evaluation code, the synthetic task data, and draft manuscript text were produced with the assistance of an AI model (Claude, Anthropic) and were reviewed, tested and edited by the authors, who take full responsibility for the content. [Note: Check the target venue's disclosure policy and adjust the wording.]

Author Contributions

[Note: Each co-author should confirm their own roles before submission.] **M. I. Sanjamala:** conceptualization, methodology, software, investigation, formal analysis, writing (original draft). **K. Prem Kumar:** investigation (Apple Silicon runs), validation (human rating), writing (review and editing). **R. Smith:** conceptualization (institutional context), adjudication of ratings, writing (review and editing). **A. Singh:** supervision, methodology, writing (review and editing).

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